



CSE 3433- SOFTWARE ARCHITECTURE

Lecturer: Dr Mohamad Nor Bin Hasan

Topic: Individual Log's Book Event-Driven Job Recruitment &  
Application Processing System (Event-Driven Architecture)

Group: Group 12

Name: Muhammad Syukran Bin Mohd Zakri

Matrics Number: S75572

Team Members:

| No | Name                                     | Matric Number |
|----|------------------------------------------|---------------|
| 1  | MUHAMAD NUR IMAN HAFIZI BIN MOHD<br>ADIB | S75659        |
| 2  | MUHAMMAD HARITH RIEZHAN BIN SYANAZ       | S75804        |
| 3  | MUHAMMAD SYUKRAN BIN MOHD ZAKRI          | S75572        |

## 1. Project Background

The Event-Driven Job Recruitment & Application Processing System is a web-based recruitment platform developed to improve the efficiency of job searching and application management. The system provides a centralized platform where candidates can search for available job opportunities, submit applications, and monitor their application progress, while recruiters can manage job postings and review candidate applications.

Throughout this project, I was assigned as the Frontend Lead. My responsibility focused on designing and developing the user interface, implementing frontend functionality, and ensuring that the frontend components properly interact with backend services.

The proposed system consists of several components including job management, application processing, notification handling, and data management. The frontend application acts as the main interaction layer where candidates and recruiters communicate with the system.

### 1.2 My Role and Responsibility

As the Frontend Lead, these are my responsibility was managing development of user interface layer for the recruitment system. For the main responsibilities as Frontend Lead is designing overall interface structure which for candidates and recruiters, developing JSP-based web pages, managing navigation between each modules system, connecting form with Java Servlet Controller, testing functionality and also improving user experience for consistency purpose.

My role required continuous communication with backend developers to ensure that frontend requests, data display, and system workflow were aligned with the overall architecture. The frontend development was implemented using JSP, HTML, CSS, and JavaScript. JSP was selected because it allows dynamic web page generation and integration with Java Servlet backend components.

## 2. Weekly Progress

### 2.1 Phase 1- Project Proposal

|                    |                                                                                                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Activities | Discuss project idea by giving each role member what they should do, analysed functional and non-functional requirement of user and discussing to design frontend based on system objective. |
| Contributions      | writing the problem description and system objectives sections of the proposal and contribute by identifying the required user interface and user flow for system.                           |
| Challenge Faced    | The team are hardly to decide about recruitment feature such as interview scheduling and analytics of the users.                                                                             |
| Solution           | Discussing and select core features that could be implemented to functional boundaries which is Job Management, Candidate Management, Job Search/Discovery and Notification dispatches.      |

### 2.2 Phase 2- Architectural Design

|                    |                                                                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Project Activities | Design and mapped out abstract system architecture pattern of frontend and analysed frontend communication with Event-Driven Architecture. |
| Contributions      | The frontend designed as main interaction layer where candidates and recruiters communicate with backend services through API Gateway.     |
| Challenge Faced    | Confusion regarding search queries occurred where frontend should make directly to interface Search and Index Service.                     |
| Solution           | Frontend routes search queries directly to EventDispatcher service via gateway.                                                            |

### 2.3 Project Observation 1

|                    |                                                                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Activities | Start with development prototype by set up the Apache Netbeans project structure with MySQL. Configured the MySQL database and established the JDBC connection from the backend. |
| Contributions      | Created Web Application targets to ensure all structural assets, package dependencies and client environment variables were checked before deploying local files.                |
| Challenge Faced    | Running the NetBeans web execution server produced blocking when JSP tried to post in HTTP to backend gateway.                                                                   |

|          |                                                                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Solution | Frontend and backend lead are declared to allow developer with routing logic, establish end-to-end JSP to make sure the backend gateway are successfully worked. |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 2.4 Phase 3- Prototype Progress

|                    |                                                                                                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Activities | Frontend view bindings and error parsing. Developed the candidate job discovery workspace and focused on implementing the interface of the recruitment system using JSP, HTML, CSS and JavaScript. |
| Contributions      | Design interface toggle for button engine that lock the application submit button immediately to make user interaction within the system.                                                          |
| Challenge Faced    | When sending candidate applications, the UI suffered severe lag because the network layer waited synchronously for deep background checks to complete.                                             |
| Solution           | Double check frontend user and define problem that causing the UI network handler to catch instant from the gateway.                                                                               |

#### 2.5 Project Observation 2

|                    |                                                                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Activities | Conducted again the frontend system to ensure elements correctly mirrored with architecture microservice pipeline events and also tracking accuracy active of workspaces of the candidates and recruiters. |
| Contributions      | Evaluated UI notification indicators when state alerts were published to Apache Netbeans. Fixed rendering bugs in the candidate tracking interface where hiring status movements failed to refresh.        |
| Challenge Faced    | Presentation views required hard page refreshes which degraded user experience required by our non-functional targets.                                                                                     |
| Solution           | Fix the coding by making automated data cycles within presentation state that ensure state tracking display real-time updates of event.                                                                    |

#### 2.6 Phase 4- Project Report & Individual Logbook

|                    |                                                                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Activities | Compiled all project report architecture which cover design, functional and non-functional requirements, technology stack, service and component identification, system diagrams, team contributions, and conclusion. |
| Contributions      | Ensured all tool names, class names, and table names in the report accurate and consistent with the actual implemented codebase. Define all the                                                                       |

|                 |                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
|                 | component required sections are in the report with correct format.                                              |
| Challenge Faced | Ensuring the report accurately reflected the final implemented system rather than the originally planned design |
| Solution        | Cross-referenced every report section against the actual codebase to verify accuracy.                           |

### 2.7 Phase 5- Final Demonstration

|                    |                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Activities | Ensured all three cloud services (HTML5 / CSS3 / JavaScript) were running and responsive.                                                                             |
| Contributions      | Define explanations of the layered architecture, separation of concerns and role of each tool in the deployment stack that been used.                                 |
| Challenge Faced    | Potential network drops during local execution could cause noticeable lag when rendering distributed records from cloud-hosted databases.                             |
| Solution           | Pre-warmed data connections in the Apache NetBeans workspace before the presentation, ensuring quick database access and smooth rendering during the live evaluation. |

## 3. Generative AI Usage

| Tools Used | Purpose                                                                                                    | Prompt Used                                                                                                                      | Output                                                                                                                                     | How it was used                                                                                                                                     |
|------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Claude.ai  | Generate a UML-compliant deployment diagram for the EDA system architecture                                | Generate a UML deployment diagram suitable for this system using HTML5 / CSS3 / JavaScript and Cloudinary API                    | Generated a UML deployment diagram with correct stereotypes including executionEnvironment, component, artifact, database, and table.      | Referring to the generated output, the final diagram was designed using VisualParadigm.                                                             |
| Claude.ai  | To design a dynamic dashboard interface for the Recruiter role to effectively view, filter, and transition | As a Frontend Lead, I need to design a clean, minimalist recruiter dashboard component in React. It should display a grid of job | Provided a highly modular React component structure styled with clean, minimalist CSS guidelines. It delivered optimized hooks logic using | he modular grid layout and the color-coded application status tags provided by Claude were adopted as the foundational blueprint for developing the |

|         |                                                                                                                                                 |                                                                                                                                                                                                                                  |                                                                                                                                                                                              |                                                                                                                                                                |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | incoming candidate applications                                                                                                                 | applications, include visual tags for status changes (Pending, Shortlisted, Rejected), and handle state transitions efficiently using React Hooks.                                                                               | useState and useEffect to capture asynchronous state transitions when managing incoming application pipelines.                                                                               | recruiter-workspace.jsp within the Apache NetBeans workspace.                                                                                                  |
| Gemini  | To understand how to properly customize Apache Ant targets in NetBeans (build.xml) to run automated checks before bundling frontend components. | I am using Apache NetBeans for my project which has a build.xml file. How can I use the -pre-compile and -post-compile targets to echo build messages and verify that my local asset directories exist before the project runs?" | Provided structural XML syntax demonstrating how to override empty target blocks by embedding <echo> commands and <available> property verification blocks natively inside the build script. | Copied and integrated the Web structure into our system's build.xml configuration, automating environment validation within NetBeans before local compilation. |
| ChatGPT | To identify possible causes of JSP form submission problems.                                                                                    | My JSP form is not passing data correctly to Java Servlet. What should I check?                                                                                                                                                  | The response explained that form submission issues may occur due to incorrect configuration between the frontend page and backend controller.                                                | The suggestions were used as a debugging checklist when checking the connection between JSP pages and Java Servlet controllers.                                |

#### 4. Reflection and Learning outcome

Throughout this project, I gained valuable experience in frontend development within a software architecture environment.

Initially, I viewed frontend development mainly as designing user interfaces. However, this project helped me understand that frontend development involves more than visual design. A frontend developer needs to understand user requirements, backend communication, data flow and system architecture to be included due to the request of system's user.

Working as the Frontend Lead improved my technical skills in:

- JSP-based web development.
- Interface design.
- Frontend and backend integration.
- Debugging user interface problems.
- Software documentation.

The project also improved my teamwork skills because frontend development required continuous communication with backend and database team members.

Through this experience, I developed a better understanding of how individual system components work together to create a complete software solution.

#### 5. Academic Integrity Declaration

I hereby declare that this individual logbook represents my actual contribution and learning experience throughout the development of the Event-Driven Job Recruitment & Application Processing System.

My contribution focused on frontend development activities including user interface design, JSP page development, frontend testing, integration with backend components, and documentation preparation.

All information recorded in this logbook is based on my own involvement and understanding throughout the project development process.

Generative AI tools were used only as a supporting resource for learning, idea generation, debugging guidance, and improving technical explanations. All generated suggestions were reviewed and modified according to the actual system implementation.

I acknowledge the importance of academic integrity and take responsibility for all work documented in this individual logbook.

|               |                                 |
|---------------|---------------------------------|
| Name          | Muhammad Syukran Bin Mohd Zakri |
| Matric Number | S75572                          |
| Group Number  | 12                              |
| Role          | Frontend Lead                   |